

Combinatorics, Computer Science, and Permutation Patterns

William Turner

Department of Mathematics & Computer Science

3 October 2017



Outline

- 1 Combinatorics
- 2 Permutation Patterns
- 3 Computer Science Application
- 4 Generating Functions
- 5 Binomial Theorem

Combinatorics is concerned with the study of...

- arrangements
- patterns
- designs
- assignments
- schedules
- connections
- configurations

Combinatorical Problems

- A computer scientist considers *patterns* of digits and switches to encode complicated statements.
- A shop supervisor prepares *assignments* of workers to tools or to work areas.
- An agronomist *assigns* test crops to particular fields.
- An electrical engineer considers alternative *configurations* for a circuit.
- An industrial engineer considers alternative production *schedules* and workplace *configurations* to maximize efficient production.
- A university scheduling officer *arranges* class meeting times and students' *schedules*.
- A chemist considers possible *connections* between various atoms and molecules and *arrangements* of atoms into molecules.
- A transportation officer *arranges* bus or plane *schedules*.
- A linguist considers *arrangements* of words in unknown alphabets.

Three Basic Problems

Existence: Is there at least one arrangement of a particular kind?

Counting: How many arrangements are there?

Optimization: Among all possible arrangements, what is the best one according to some criterion?

Basic Counting Rules

The Sum Rule

If one event can occur in n_1 ways and a second event in n_2 different ways, there are $n_1 + n_2$ ways in which either the first event or the second event can occur, but not both.

Example

I have five apples and four oranges. How many ways can I select one piece of fruit—either an apple or an orange?

$$5 + 4 = 9$$

Basic Counting Rules

The Product Rule

If one event can occur in n_1 ways and a second event in n_2 different ways, independent of the first, then there are $n_1 \cdot n_2$ ways in which both the first event and the second event can occur.

Example

I have five apples and four oranges. How many ways can I select one apple and one orange?

$$5 \cdot 4 = 20$$

- A *permutation* is an arrangement of the numbers in the set

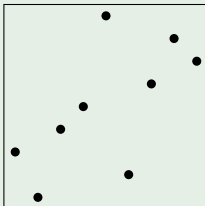
$$[n] = \{1, 2, 3, \dots, n\}$$

for some $n > 0$.

- The length of permutation σ is denoted $|\sigma|$.
- S_n is the set of all permutations of length n .
- Common to plot permutations graphically by points $(i, \sigma(i))$ where $\sigma = \sigma(1) \dots \sigma(n)$.

Example

Consider $\sigma = 314592687 \in S_9$.



Counting Permutations

How many permutations are there of length n ?

There are n ways to choose the first element, $n - 1$ ways to choose the second, ... , $n - i$ ways to choose the $i + 1$ -st element, ..., 1 way to choose the n th element:

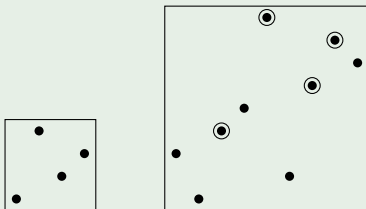
$$|S_n| = n(n - 1)(n - 2) \cdots (n - i) \cdots 1 = n!$$

Permutation Pattern

- Consider some subsequence (not necessarily contiguous) of σ .
- Replace i -smallest entry by i (Reduce their values)
- This forms a *pattern* π contained in σ ($\pi \leq \sigma$)

Example

$1423 \leq 314592687$ because 4968 (among others) is ordered the same way:

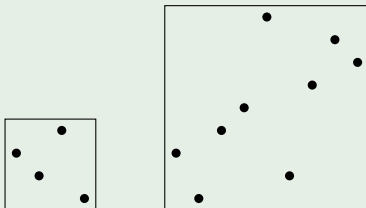


Pattern Avoidance

- σ avoids π if $\pi \not\leq \sigma$
- $\sigma \in S_n(\pi)$

Example

$314592687 \in S_9(3241)$ because $3241 \not\leq 314592687$:

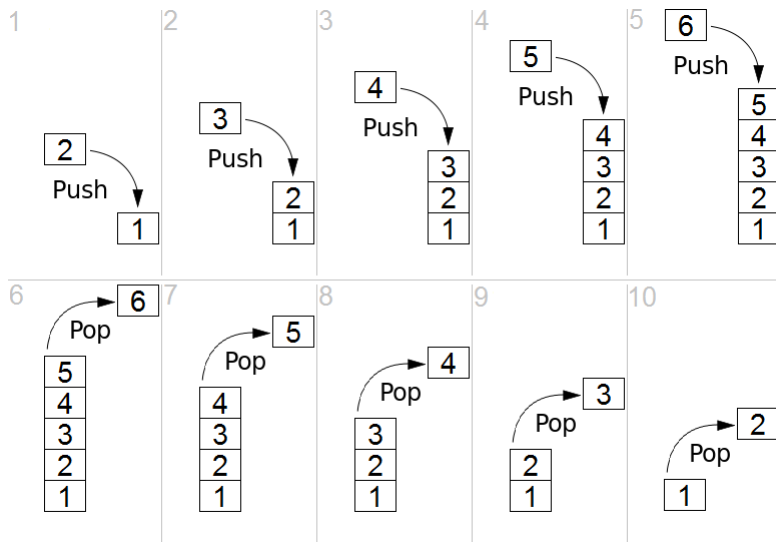


- Computer scientists want to know how to quickly sort a list (permutation).
- Knuth (1968): What permutations can be sorted by an algorithm with a single stack as storage?
- Linear time algorithm ($2n$ operations — fast!)

Stack

- Linear structure with insertion and deletion on same end
- “Push” and “pop”
- Last in, first out (LIFO)
- E.g., stack of plates in cafeteria

Stack Example



Sorting with a Stack

Knuth's algorithm

- Initialize an empty stack
- For each input value x :
 - While the stack is nonempty and x is larger than the top item on the stack, pop the stack to the output
 - Push x onto the stack
- While the stack is nonempty, pop it to the output

Example (4132)

- | | |
|----------------------|----------------------|
| ① Push 4 onto stack | ⑤ Push 2 onto stack |
| ② Push 1 onto stack | ⑥ Pop 2 off of stack |
| ③ Pop 1 off of stack | ⑦ Pop 3 off of stack |
| ④ Push 3 onto stack | ⑧ Pop 4 off of stack |

Sorting with a Stack

Example (3142)

- | | |
|----------------------|----------------------|
| ➊ Push 3 onto stack | ➋ Push 4 onto stack |
| ➌ Push 1 onto stack | ➍ Push 2 onto stack |
| ➎ Pop 1 off of stack | ➏ Pop 2 off of stack |
| ➐ Pop 3 off of stack | ➑ Pop 4 off of stack |

- This produces 1324
- The problem is the 342, or the pattern 231



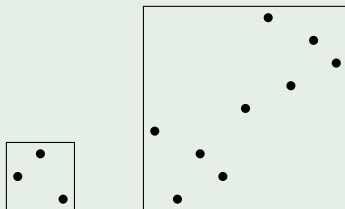
Stack Sortable

Theorem

A permutation σ is stack sortable if and only if $\sigma \in S_n(231)$.

Example

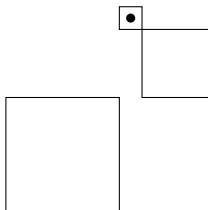
The permutation 413259687 is stack sortable.



Counting Stack Sortable Permutations

How big is $|S_n(231)|$?

$\sigma \in S_n(231)$ if and only if σ has recursive structure



If $|S_n(231)| = c_n$, then $c_n = \sum_{i=1}^n c_{i-1}c_{n-i}$ for $n \geq 1$ and $c_0 = 1$.

Counting Stack Sortable Permutations

- $c_0 = 1$
 - The empty permutation
- $c_1 = c_0 c_0 = 1$
 - The only permutation of length 1
 - 1
- $c_2 = c_0 c_1 + c_1 c_0 = 1 \cdot 1 + 1 \cdot 1 = 2$
 - All permutations of length 2
 - 12, 21
- $c_3 = c_0 c_2 + c_1 c_1 + c_2 c_0 = 1 \cdot 2 + 1 \cdot 1 + 2 \cdot 1 = 5$
 - All permutations of length 3 except 231
 - 123, 132, 213, 312, 321
- $c_4 = c_0 c_3 + c_1 c_2 + c_2 c_1 + c_3 c_0 = 1 \cdot 5 + 1 \cdot 2 + 2 \cdot 1 + 1 \cdot 1 = 14$
- $c_5 = c_0 c_4 + c_1 c_3 + c_2 c_2 + c_3 c_1 + c_4 c_0 = 42$
- $c_6 = c_0 c_5 + c_1 c_4 + c_2 c_3 + c_3 c_2 + c_4 c_1 + c_5 c_0 = 132$

Closed formula for c_n ?

Generating Functions

Generating Function

The (ordinary) generating function for the sequence (a_n) is defined to be the power series

$$G(x) = \sum_{k=0}^{\infty} a_k x^k = a_0 x^0 + a_1 x^1 + a_2 x^2 + \cdots$$

Example (Binomial Theorem)

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = \binom{n}{0} + \binom{n}{1} x + \binom{n}{2} x^2 + \cdots + \binom{n}{n} x^n$$

Binomial Theorem

Consider $f(x) = (1+x)^u$ for $u \in \mathbb{R}$.

- $f'(x) = u(1+x)^{u-1}$
- $f''(x) = u(u-1)(1+x)^{u-2}$
- $f^{(r)}(x) = u(u-1) \cdots (u-r+1)(1+x)^{u-r}$

$$\begin{aligned}(1+x)^u &= 1 + ux + \frac{u(u-1)}{2!}x^2 + \cdots + \frac{u(u-1) \cdots (u-r+1)}{r!}x^r + \cdots \\ &= \sum_{r=0}^{\infty} \binom{u}{r} x^r\end{aligned}$$

where

$$\binom{u}{r} = \begin{cases} \frac{u(u-1) \cdots (u-r+1)}{r!}, & \text{if } r > 0 \\ 1 & \text{if } r = 0 \end{cases}$$

Airthmetic on Generating Functions

Let $A(x) = \sum_{n=0}^{\infty} a_n x^n$ and $B(x) = \sum_{n=0}^{\infty} b_n x^n$.

$$A(x) + B(x) = \sum_{n=0}^{\infty} (a_n + b_n) x^n$$

$$\begin{aligned} A(x)B(x) &= (a_0 + a_1x + a_2x^2 + \dots)(b_0 + b_1x + b_2x^2 \dots) \\ &= a_0b_0 + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2 + \dots \\ &= \sum_{n=0}^{\infty} \left(\sum_{i=0}^n a_i b_{n-i} \right) x^n \end{aligned}$$

Counting Stack Sortable Permutations

Consider generating function for $c_n = |S_n(231)|$:

$$\begin{aligned}C(x) &= \sum_{n=0}^{\infty} c_n x^n = c_0 + \sum_{n=1}^{\infty} \left(\sum_{i=1}^n c_{i-1} c_{n-i} \right) x^n \\&= c_0 + c_0 c_0 x + (c_0 c_1 + c_1 c_0) x^2 + (c_0 c_2 + c_1 c_1 c_2 c_0) x^3 + \cdots \\&= 1 + x \left(c_0 c_0 + (c_0 c_1 + c_1 c_0) x + (c_0 c_2 + c_1 c_1 c_2 c_0) x^2 + \cdots \right) \\&= 1 + x (c_0 + c_1 x + c_2 x^2 + \cdots)^2 = 1 + x C(x)^2\end{aligned}$$

Thus,

$$x C(x)^2 - C(x) + 1 = 0$$

and the quadratic formula gives

$$C(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}$$

Counting Stack Sortable Permutations

Applying the Binomial Theorem:

$$\sqrt{1-4x} = (1-4x)^{1/2} = \sum_{n=0}^{\infty} \binom{1/2}{n} (-4x)^n = \sum_{n=0}^{\infty} \binom{1/2}{n} (-4)^n x^n$$

For $n \geq 1$,

$$\begin{aligned} \binom{1/2}{n} (-4)^n &= \frac{\left(\frac{1}{2}\right) \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \cdots \left(-\frac{2n-3}{2}\right)}{n!} (-4)^n \\ &= \frac{-2^n (1 \cdot 1 \cdot 3 \cdot 5 \cdots (2n-3))}{n!} \\ &= -\frac{2}{n} \cdot \frac{2^{n-1}}{(n-1)!} (1 \cdot 1 \cdot 3 \cdot 5 \cdots (2n-3)) \frac{(n-1)!}{(n-1)!} \\ &= -\frac{2}{n} \cdot \frac{(1 \cdot 1 \cdot 3 \cdot 5 \cdots (2n-3))(2 \cdot 4 \cdot 6 \cdots (2n-2))}{(n-1)! (n-1)!} \\ &= -\frac{2}{n} \cdot \frac{(2n-2)!}{(n-1)! (n-1)!} = -\frac{2}{n} \binom{2n-2}{n-1} \end{aligned}$$

Counting Stack Sortable Permutations

Thus,

$$\sqrt{1-4x} = (1-4x)^{1/2} = 1 - \sum_{n=1}^{\infty} \frac{2}{n} \binom{2n-2}{n-1} x^n$$

Two roots:

$$C_1(x) = \frac{1 + \sqrt{1-4x}}{2x}$$

and

$$C_2(x) = \frac{1 - \sqrt{1-4x}}{2x}$$

Counting Stack Sortable Permutations

$$\begin{aligned}C_1(x) &= \frac{1 + \sqrt{1 - 4x}}{2x} = \frac{1}{2x} \left(1 + \left(1 - \sum_{n=1}^{\infty} \frac{2}{n} \binom{2n-2}{n-1} x^n \right) \right) \\&= \frac{1}{2x} \left(2 - \sum_{n=1}^{\infty} \frac{2}{n} \binom{2n-2}{n-1} x^n \right) \\&= \frac{1}{x} - \sum_{n=1}^{\infty} \frac{1}{n} \binom{2n-2}{n-1} x^{n-1} = \frac{1}{x} - \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n\end{aligned}$$

If $n \geq 1$,

$$c_n = -\frac{1}{n+1} \binom{2n}{n} < 0,$$

which is impossible.

Counting Stack Sortable Permutations

$$\begin{aligned}C_2(x) &= \frac{1 - \sqrt{1 - 4x}}{2x} = \frac{1}{2x} \left(1 - \left(1 - \sum_{n=1}^{\infty} \frac{2}{n} \binom{2n-2}{n-1} x^n \right) \right) \\&= \frac{1}{2x} \sum_{n=1}^{\infty} \frac{2}{n} \binom{2n-2}{n-1} x^n = \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n\end{aligned}$$

If $n \geq 1$,

$$c_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!} = C_n,$$

the *Catalan numbers*, which count many combinatorial objects:

- [Wikipedia](#)
- [The Online Encyclopedia of Integer Sequences](#)